

Appendix: Nonparametric Bounds for Binary Outcomes

Obtaining meaningful nonparametric bounds becomes more challenging when the outcome measure is binary rather than continuous. Binary outcomes contain little information to begin with, and the observed outcomes do not furnish direct information about the joint distribution of $Y_i(1)$ and $Y_i(0)$. Nevertheless, when working with binary data, it is often interesting to apply bounds to the joint distribution of $Y_i(1)$ and $Y_i(0)$ in order to get a sense of the range of possible underlying parameters.

[insert Box A9.1 here]

Consider, for example, the discrimination experiment reported by Stewart Page (1998). Page sought to test whether disclosing that one is gay while inquiring by phone about the availability of an apartment changed the probability that the apartment was described as still available. The hypothesis is that apartment managers in Canada and the US would be less likely to describe the apartment as available if they believed the caller to be gay. For ease of exposition, we restrict our attention to outcomes involving male callers. Page found that in 30 out of 45 calls made in the control condition, callers were told that the apartment was still available: $\widehat{\Pr}(Y_i(0) = 1) = .667$. When callers disclosed that they were gay, just 10 of 45 apartments were described as still available: $\widehat{\Pr}(Y_i(1) = 1) = .222$. Based on these marginal distributions, what inferences can we draw about the joint distribution of $Y_i(1)$ and $Y_i(0)$, depicted in Table A9.1?

[insert Table A9.1 about here]

Applying the bounds from Box A9.1 to the Page experiment generates ranges of possible values for each cell, indicated by [low, high] entries in Table A9.2. The wide range of possible values in each cell is a bit disappointing (and the bounds would be wider still if we took sampling variability into account), but we nonetheless learn something about the latent types of apartment managers. The cell in which both $Y_i(1)$ and $Y_i(0)$ are 1 indicates that no more than 22% of managers were prepared to rent to anyone regardless of whether they disclosed their sexual orientation. The cell in which $Y_i(1) = 0$ and $Y_i(0) = 1$ indicates that between 44% and 67% of managers discriminated against callers who disclosed a gay sexual orientation. The upper left cell suggests that no more than one-third of the apartments were truly unavailable regardless of who inquired about them. These nonparametric bounds are less helpful when it comes to understanding the amount of heterogeneity in the treatment effect. The lower bound of 44% tells us that almost half of the managers discriminated against callers who disclosed a gay sexual orientation, but anywhere from 0% to 22% of managers may have discriminated *in favor* of putatively gay callers.

[insert Table A9.2 about here]

Uncertainty about heterogeneous treatment effects is not specific to this example. When applied to binary outcomes, nonparametric bounds rarely establish that a treatment only moves subjects in one direction. The bounds become narrow as either $\Pr(Y_i(1) = 1)$ or $\Pr(Y_i(0) = 1)$ approach the extremes of either zero or one. For example, if $\Pr(Y_i(0) = 0) = .667$ and $\Pr(Y_i(1) = 1) = .05$, we would know that between 45% and 50% of managers discriminated against gay men and between 0% and 5% discriminated in favor of them. When $\Pr(Y_i(1) = 1) = .01$, the bounds narrow such that at most 1% of

the managers discriminated in favor of putatively gay callers. But such extreme examples are rare in practice. Typically, researchers find themselves unable to rule out the possibility that large proportions of the sample responded to the treatment in opposite directions.

[insert Box A9.2 here]

One way to proceed is to bring additional assumptions to bear. The literature on “attributable effects” investigates the inferences that can be drawn if one stipulates that the treatment moves everyone in the same direction if it moves them at all. In the case of binary outcomes, this assumption solves the identification problem. When we rule out discrimination in favor of putatively gay callers, the joint distribution of potential outcomes is revealed. The results are shown in Table A9.3.

[insert Table A9.3 about here]

Of course, the notion that the treatment only moves managers in one direction is an assumption, not an empirical result. For this assumption to be convincing, it needs to be backed up by some kind of evidence. Is there descriptive information suggesting that managers harbor uniformly negative stereotypes about gay men in general and gay tenants in particular? If the experiment were replicated on a large scale with a diverse array of apartment managers, would we find that certain subgroups of managers declare that apartments are available at higher rates when putatively gay men make inquiries?

Bounds provide some hints about the extent of heterogeneity, but unless we bring in outside assumptions (e.g., $\tau_i \geq 0$), bounds will often give us only a vague sense of the amount of treatment heterogeneity. When outcomes are continuous rather than binary,

and when untreated potential outcomes have low variance in relation to the variance in treatment effects, bounds may provide useful guidance. When they suggest that heterogeneity exists, bounds direct further investigation into the systematic causes of varying treatment effects. We now discuss two general approaches.

Table A9.1: Illustration of the Problem of Inferring the Joint Distribution of $Y_i(1)$ and $Y_i(0)$ from the Marginal Distributions of Each Variable, Using Data from Page's (1998) Study of Housing Discrimination

	$Y_i(0)$			
$Y_i(1)$		0	1	
	0	?	?	.778
	1	?	?	.222
		.333	.667	N=90

Table A9.2: Nonparametric Bounds on the Joint Distribution of $Y_i(1)$ and $Y_i(0)$ Based on the Marginal Distributions of Each Variable, Using Data from Page's (1998) Study of Housing Discrimination

	$Y_i(0)$			
$Y_i(1)$		0	1	
	0	[.111,.333]	[.444,.667]	.778
	1	[0,.222]	[0,.222]	.222
		.333	.667	N=90

Note: The bounds reported in this table do not account for sampling variability in the observed marginal distributions.

Table A9.3: Nonparametric Identification of the Joint Distribution of $Y_i(1)$ and $Y_i(0)$
Assuming “Attributable Effects,” Using Data from Page’s (1998) Study of Housing
Discrimination

	$Y_i(0)$			
$Y_i(1)$		0	1	
	0	.333	.444	.778
	1	0	.222	.222
		.333	.667	

Box A9.1**Nonparametric Bounds for 2x2 Tables**

Heckman, Smith, and Clements (1997, pp.501-503) work out the bounds for any 2x2 table given its marginal distribution. The bounds for each cell in a 2x2 table are as follows:

$$\begin{aligned} \max[\Pr(Y_i(0) = 1) + \Pr(Y_i(1) = 1) - 1, 0] &\leq \Pr(Y_i(0) = 1, Y_i(1) = 1) \\ &\leq \min[\Pr(Y_i(0) = 1), \Pr(Y_i(1) = 1)] \end{aligned}$$

$$\begin{aligned} \max[\Pr(Y_i(1) = 1) - \Pr(Y_i(0) = 1), 0] &\leq \Pr(Y_i(0) = 0, Y_i(1) = 1) \\ &\leq \min[1 - \Pr(Y_i(0) = 1), \Pr(Y_i(1) = 1)] \end{aligned}$$

$$\begin{aligned} \max[\Pr(Y_i(0) = 1) - \Pr(Y_i(1) = 1), 0] &\leq \Pr(Y_i(0) = 1, Y_i(1) = 0) \Pr(Y_i(0) = 1) \\ &\leq \min[\Pr(Y_i(0) = 1), 1 - \Pr(Y_i(1) = 1)] \end{aligned}$$

$$\begin{aligned} \max[1 - \Pr(Y_i(0) = 1) - \Pr(Y_i(1) = 1), 0] &\leq \Pr(Y_i(0) = 0, Y_i(1) = 0) \\ &\leq \min[1 - \Pr(Y_i(0) = 1), 1 - \Pr(Y_i(1) = 1)] \end{aligned}$$